Metric random tilings: exact entropy and the diffraction of counted disorder

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Tiling mathematics has twice become the language of a state of matter: crystals, then quasicrystals, whose thermodynamics are described by the random tiling of fixed shapes. There the counted variable is the arrangement rather than the geometry, although the metric is what chemical design sets and diffraction measures. Here we introduce metric random tilings, periodic networks of locked topology whose edge lengths fluctuate under exact face closure, so that the geometry itself is countable. Purpose-built counting engines enumerate Kepler's eleven uniform networks and divide them into seven zero-entropy ones and four carriers with positive metric entropy, each certified by an explicit construction. On the honeycomb, the densest carrier, all edge-length value domains considered carry extensive entropy, with the arithmetic class reaching $s = 0.544\,k_B$ per tile (fit band 0.53–0.58), the other two ≈ 0.2. In diffraction, translational increase of the value domain leaves the entropy exactly invariant while the coherent fraction of the Bragg intensity at fixed patch size varies from 9×10^{-4} to 0.58, following the Debye–Waller law of the amplitude flow. Metric disorder thus enters the exact-counting paradigm, and is identifiable in real materials by Bragg spots wrapped in metric halos.

INTRODUCTION

Counting disorder exactly is the strongest statement statistical mechanics can make about a phase, and tiling models are where that statement has twice been made about real matter. Periodic tilings gave crystals their symmetry; Penrose's rhombi[1, 2] were already on the page when quasicrystalline diffraction arrived,[3–5] and random tilings of fixed shapes then supplied the quasicrystal's thermodynamics: an exactly counted configurational entropy that stabilises the phase,[6–8] and phason elasticity whose entropic origin was settled early[9, 10] and whose constants are read out of the diffuse scattering around the Bragg peaks.[11–13] Ice counts proton placements on a frozen lattice,[14] dimer and lozenge models count matchings on a fixed graph, and random tilings count arrangements of tiles whose shapes and edge lengths are fixed once and for all, including the exactly solved square-triangle and octagonal tilings.[15–17] In that tradition the metric itself has never been the counted variable: length fluctuation appears either as positional jitter or through continuously fluctuating edge lengths on a triangulation of variable connectivity,[18] never as a finite value domain under per-face closure.

This paper asks what happens when the metric is. Keep the topology completely locked, so that every vertex looks alike and every face is a fixed polygon, and let the edge lengths fluctuate under a single constraint: every face must close (Fig. 1a,b). The result is a Wang-type ensemble whose matching colour is the edge length itself,[19] and, we show, one of positive entropy density. It is the missing row of the tiling dictionary, between "fixed shapes, random arrangement" and "no rules at all", and it asks a sharp question: when does metric disorder carry extensive entropy, and what does a metric entropy do to diffraction? The arena of the question is old. The eleven uniform tilings of the plane were all drawn in Kepler's *Harmonies Mundi* of 1619,[20] and the question is asked here for every one of the eleven (Table S2).

What makes the question hard is the word exactly. The closure conditions confine every assignment to an integer lattice inside a homogeneous linear system: countable in principle. Statistical approaches to fluctuating lengths, the paracrystal being the classical one,[21] give up exact counting by construction: a continuous lattice-vector distribution carries neither a motif library nor matching rules, and fixes only an average local orientation. The constraint-counting tradition[22–24] computes the dimension of the metric freedom of a framework, never the number of integer assignments that realize it; where such integer points are counted, it is as vector partition functions of the right-hand side,[25, 26] whose dependence on the data is quasi-polynomial,[27] and none of it is an entropy over a value domain. Even the exact traditions remain on the other side of the metric: random-tiling entropy counts arrangements at fixed shapes, and the spatial entropy of Wang tiles counts colours, not lengths.[28, 29] What brings the metric row within reach is the equiangular restriction. With internal angles fixed, face closure becomes a homogeneous linear equation, and the integer points of its solution space can be counted exactly on a torus.

This paper sets up that counting and follows it through the eleven networks. The model is minimal: a planar vertex-type network Γ; an edge alphabet $A \subset \mathbb{Z}$ whose values are at once colours, Wang matching colours and lengths; and closure, one homogeneous linear condition per face. Configurations are assignments $x \in A^E$ satisfying closure everywhere, Ω counts them, and the entropy density is $s = \lim \ln \Omega / F$. The counting is exact throughout, by enumeration and transfer matrices rather than sampling; proofs, engines and protocols are collected in

the Supplemental Material (cited as SI, with sections S1–S11). The honeycomb, the counting's thickest carrier, then carries the fuller theory, in three steps: how the value domain sets the entropy, where in the counting domain that entropy hides, and what the metric disorder does to the diffraction pattern.

What such an ensemble would mean for matter is the other half of the motivation. Chemistry could reach this row through the geometric design of rigid building blocks: the edge length is then the variable that local design sets and diffraction measures, and a countable metric disorder is one that can be looked for: its signature, Bragg spots whose coherent fraction the value domain sets, wrapped in anisotropic diffuse halos. Three theoretical questions point forward: which of the eleven networks admit the exact structure this paper exploits; how far the counting-domain arithmetic of the honeycomb extends; and what strain theory, if any, describes the fluctuations of a metric tiling.

A FOUR-MECHANISM CLASSIFICATION OF METRIC ENTROPY

Figure 1 states the positive side of the classification, which networks carry metric entropy, and Fig. 2 the negative side, the mechanisms by which the others lose it. In a classical random tiling the shapes are fixed and the arrangement fluctuates (Fig. 1a); in metric disorder the topology is fixed and the edge lengths fluctuate under closure (Fig. 1b). Both carry entropy, but what decides whether it survives is the closure system itself: over the reals, whether the constraint kernel of a network scales with the number of faces; over the integers, whether the integer points on that kernel are thick. Four mechanisms implement the verdicts (Table S3): propagation, spectra, row splitting, and rigidity and deficiency.

Of the eleven uniform networks exactly four carry metric entropy at every accessed width, drawn in Fig. 1: the honeycomb, the thickest carrier (Fig. 1b), and the three thinner ones (Fig. 1c–e): 3.4.6.4, 3.12.12 and 4.6.12. Beyond the numerical verdicts, the positive side is certified constructively: at the ternary alphabet every carrier admits a local-flip certificate, a patch with two admissible completions whose disjoint copies multiply the count, giving $s \geq \ln 2$ per 9, 24, 18 and 36 faces on the honeycomb, 3.4.6.4, 3.12.12 and 4.6.12 at every system size; the binary honeycomb is certified besides by a non-local strip construction, $s \geq (2/25)\ln 2$ at every width (SI S8.3). The four share one structural motif. In each, the closure constraints carry extensive families of dependencies localised on single faces, so the local module survives: one hexagon, one independent degree of freedom. 3.4.6.4 is the instructive near miss, because its $\mathbb{R}$ dimension is extensive ($F_{\mathrm{hex}}+3$, theorem) while a triangle-locking theorem forces every triangle edge to a single global value, compressing the bulk entropy to a leak that the asymptotic counting window puts at $s_\infty = 0.040$ (SI S7). In 3.12.12 the split rows route the entropy into an extensive mid-edge sector, the honeycomb's own edge system; in 4.6.12 they leave an extensive split lattice, strictly positive at every accessed width (SI S5.3, S8.2). Figure 1f reads the four quantitatively: on log-log axes the sub-extensive readings hug the slope-$\frac{1}{2}$ guide while the extensive carriers climb toward the slope-1 area law; the positivity of the split-row pair, which this readout cannot decide, is settled by transfer-matrix spectra (SI S8.2). The kagome completes the criterion from the other side: hexagons pinned by equilateral triangles do not suffice; the modules must remain independent.

The other seven networks lose their metric entropy; Fig. 2 shows each mechanism at work. Propagation (Fig. 2a): on the square lattice closure forces each row and each column to a single length ($\Omega = k^{L_x+L_y}$ exactly), a perimeter law with $s = 0$; the same statement elevated to strips settles the whole Penrose family, quasicrystal row included. Spectra (Fig. 2b): on the kagome the closure matrix has an exactly factorised determinant whose zero set is three families of commensurate line defects, so the kernel stays sub-extensive and $\Omega \leq k^{\dim}$. Row splitting (Fig. 2c) is the arithmetic switch of the value domain, and it cuts both ways: where the split rows exhaust the constraints, as in 4.8.8, the count collapses to $\Omega = 3^{4L}$ on $\mathbb{Z}$, and even when the library is enriched to $\mathbb{Q}(\sqrt{2})$ and the dimension revives, the count stays sub-extensive; where they do not, they leave the carriers of Fig. 1c,d. Rigidity and deficiency (Fig. 2d): the criterion's rigid end, where the kernel is zero or sub-extensive and entropy dies, is occupied by the triangular lattice ($\Omega = k$), the snub pair and the locked networks of the panel.

At matched alphabet width the comparison is sharp (Table S2): at $k = 2$, the theorem-level zeros (square, kagome, 4.8.8 on $\mathbb{Z}$) face positives that are themselves theorem-level at the ternary alphabet of every carrier and on the binary honeycomb (constructive $s \geq 0.055$, spectra reading 0.21), split-row carriers strictly positive at every accessed width ($s_2 \geq 0.186$ per face at $W = 3$, constructive 0.019), and a 3.4.6.4 leak of 0.040 (constructive 0.029). Survival and magnitude are decided by the closure structure, not by alphabet width (SI S7.3, S8.2, S8.3).

THE HONEYCOMB CARRIER: VALUE DOMAINS AND WINDING SECTORS

The honeycomb is the thickest of the four carriers, with $s = 0.544$ per tile in the arithmetic class, above every other reading (SI S6.2), and this section dissects it. Both organising questions concern boundaries: what the edge-length value domain does to the entropy, and where in

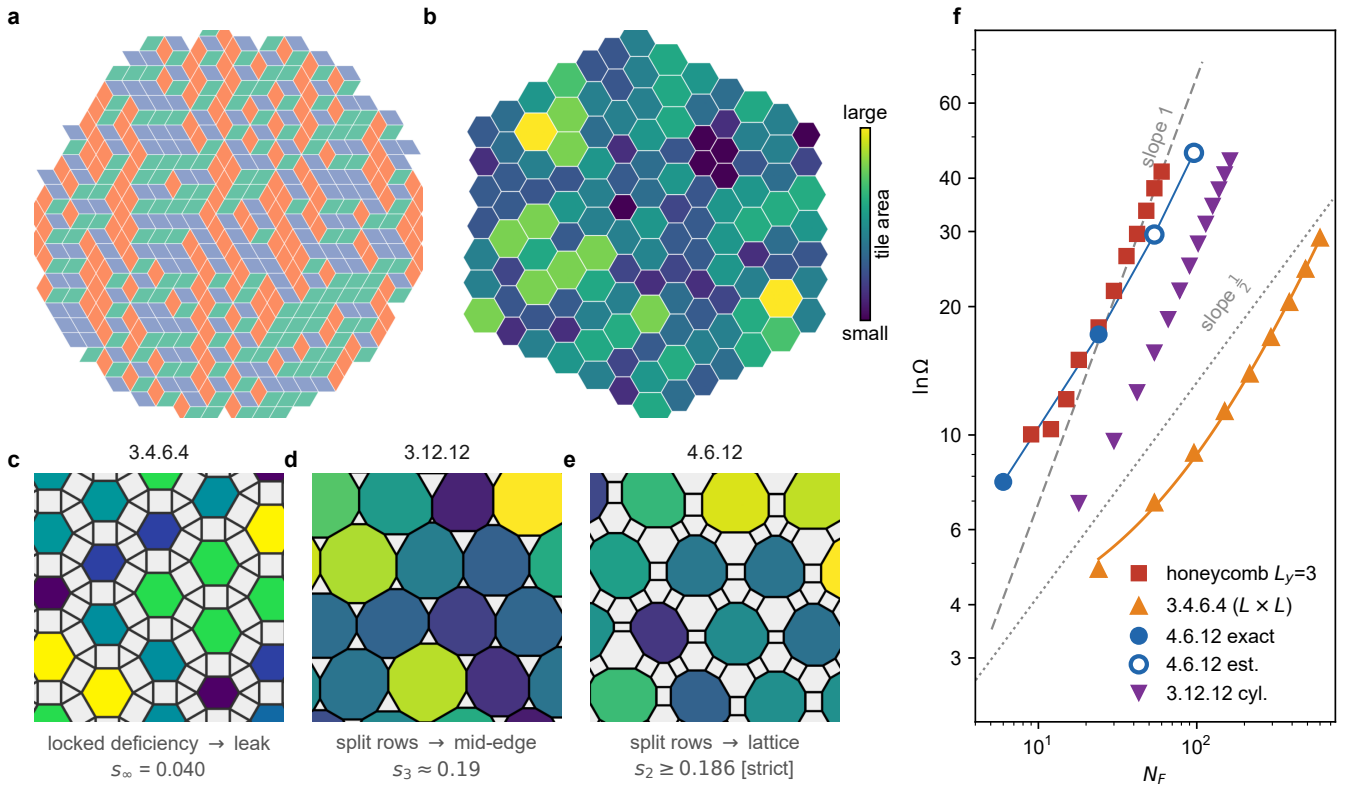


FIG. 1. Metric disorder and the four carriers of metric entropy. **a**, A classical random tiling (lozenge model): fixed shapes, random arrangement. **b**, Metric disorder on the honeycomb network: fixed topology, random edge lengths. Faces in **b**–**e** are coloured by tile area (shared colour bar, small to large, relative within each panel); edges black throughout. **c**–**e**, Counting configurations of the three thinner carriers: **c**, 3.4.6.4 ($s_\infty = 0.040$ per face); **d**, 3.12.12 ($s_3 \approx 0.19$); **e**, 4.6.12 ($s_2 \geq 0.186$ per face at $W = 3$). **f**, The carriers' quantitative readout: $\ln \Omega$ against the number of tiles N_F; filled markers, exact counts; open markers, calibrated estimates; grey guides, the area law (slope 1) and the perimeter law (slope $\frac{1}{2}$).

the counting domain, at which sizes and under which boundary conditions, that entropy hides.

What the value domain decides

The honeycomb model is "rigid angles, soft lengths": hexagons with all interior angles fixed at 120° and edge lengths drawn from a finite integer alphabet, each face required to close (two linear conditions per hexagon).[30] Because the closure conditions are homogeneous and linear, their solution space is invariant under affine maps of the alphabet (theorem; SI S6.1), so the entropy density depends only on the affine *shape class* of A. Three representatives span the alphabets studied here: the binary class {1,2}, the arithmetic class {1,2,3}, which possesses an affine midpoint, and the non-arithmetic class {1,2,4}, which does not. Their tile libraries are collected in Fig. 3b, and their counting series in Fig. 3d.

The counting domain is the torus (Fig. 3a): every edge is shared by exactly two faces, no boundary degrees of freedom remain, and the count isolates the bulk. The per-tile entropy is read along fixed-width families, $s_\infty = \lim_{L_y\to\infty} \lim_{L_x\to\infty} \ln \Omega/F$: on the binary and non-arithmetic classes the count converges to the transfer-matrix asymptotes at the resonant sizes ($L_x \equiv 0 \bmod L_y$), where the winding sectors, threads of the metric field that wrap the torus, are exposed, while the remaining sizes, with coprime periods, are pinned by an exact closed form, $\Omega = k^{L_x+L_y+1}$, bitwise at every measured size, short ($L_x \leq 20$) and long ($L_x \leq 150$) alike; the arithmetic class crosses over to its asymptote within 10–20 columns and never sits on the closed form; the rule separating the cases is the arithmetic of the counting domain and of the alphabet (SI S6.3).

The width scaling makes the verdict quantitative (Fig. 3d,e). Under a mixed global fit across all measured widths, the per-tile entropy $\lambda(L_y)/L_y$ converges to $s_\infty = 0.544$ for the arithmetic class; the binary and non-arithmetic classes converge together near 0.2, both inside the common fit band 0.17–0.27 (SI S6.2). Both asymptotes are anchored from below by constructions: flower-shaped local flips give $s \geq \ln 2/9 \approx 0.077$ for the arithmetic class at every system size, and an explicit stripe

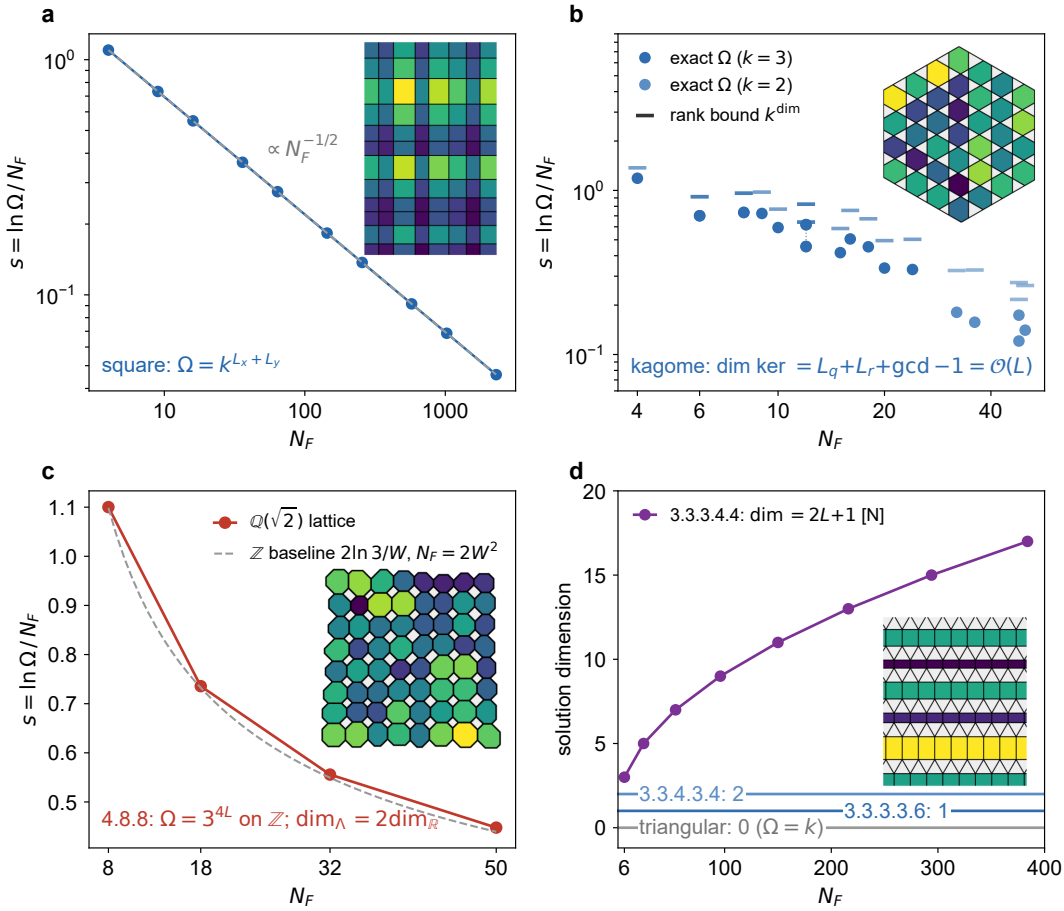


FIG. 2. Four mechanisms of metric-entropy decay. Each panel pairs a representative counting configuration (inset, same colouring as Fig. 1) with the mechanism's sub-extensive readout, plotted against the face count N_F; together the four mechanisms kill the metric entropy of the seven zero-entropy networks (Tables S2, S3). **a**, Propagation: the square lattice obeys an exact perimeter law, $s = 2\ln k/L$ (dots: exact counts, $k = 3$, $L = 2$–48; dashed guide $\propto N_F^{-1/2}$); the strip theorem extends the mechanism to the whole Penrose family. **b**, Spectra: kagome counts (dots, $k = 3$ and $k = 2$) sit beneath the decaying rank bound $k^{\dim\ker}$, full rank at infinity. **c**, Row splitting: 4.8.8 on the $\mathbb{Q}(\sqrt{2})$ library rides the $\mathbb{Z}$ baseline $2\ln 3/W$ downward, the dimension revived but the count sub-extensive; on the integer alphabet the split rows exhaust the closure, $\Omega = 3^{4L}$ exactly. **d**, Rigidity and deficiency: solution-space dimension of the locked networks (triangular, 0; the snub pair, 1 and 2; 3.3.3.4.4, $2L + 1$, sub-extensive).

construction gives $s \geq (2/25)\ln 2 \approx 0.055$ for the binary class at every width $W \geq 3$; its elementary excitations occupy entire stripes rather than local patches (SI S8.3). The separation at the level of counts needs no extrapolation: the arithmetic class outcounts the binary at every measured size, by a factor of forty-four at 3×3 (22 767 against 518). Figure 3c shows sample patches; the low-q scattering of the binary field carries the thinnest-entropy class's domain signature (SI S10.2).[31]

Along fixed-aspect-ratio commensurate families the winding-sector excess per tile rises monotonically with width and aspect ratio, tracking the class asymptotes: the commensurate families carry the entropy itself (SI S6.3); the arithmetic honeycomb's width response is shared to 3×10^{-4} by the mid-edge sector of 3.12.12: one edge system, two networks (SI S5.3). Independent of any periodic identification, direct open-boundary disk counts reproduce the class-resolved densities from the area growth, $a = 0.231(1)$ nats per face for the binary class and 0.560(4) for the arithmetic class (SI S6.3).

THE VALUE DOMAIN IN DIFFRACTION: HALOS, ANISOTROPY, AND METRIC TENSION

The metric field diffracts as a lattice with phason halos. For each value domain we draw twenty closure-respecting configurations at matched size (169 holes; SI S9) and compute exact structure factors: the ensemble-averaged $|F(q)|^2$ shows Bragg spots on the ideal shell positions. Bond-orientational order is the identity $\psi_6 \equiv 1$,[32] so the spots sit on the lattice shells for any closure-respecting

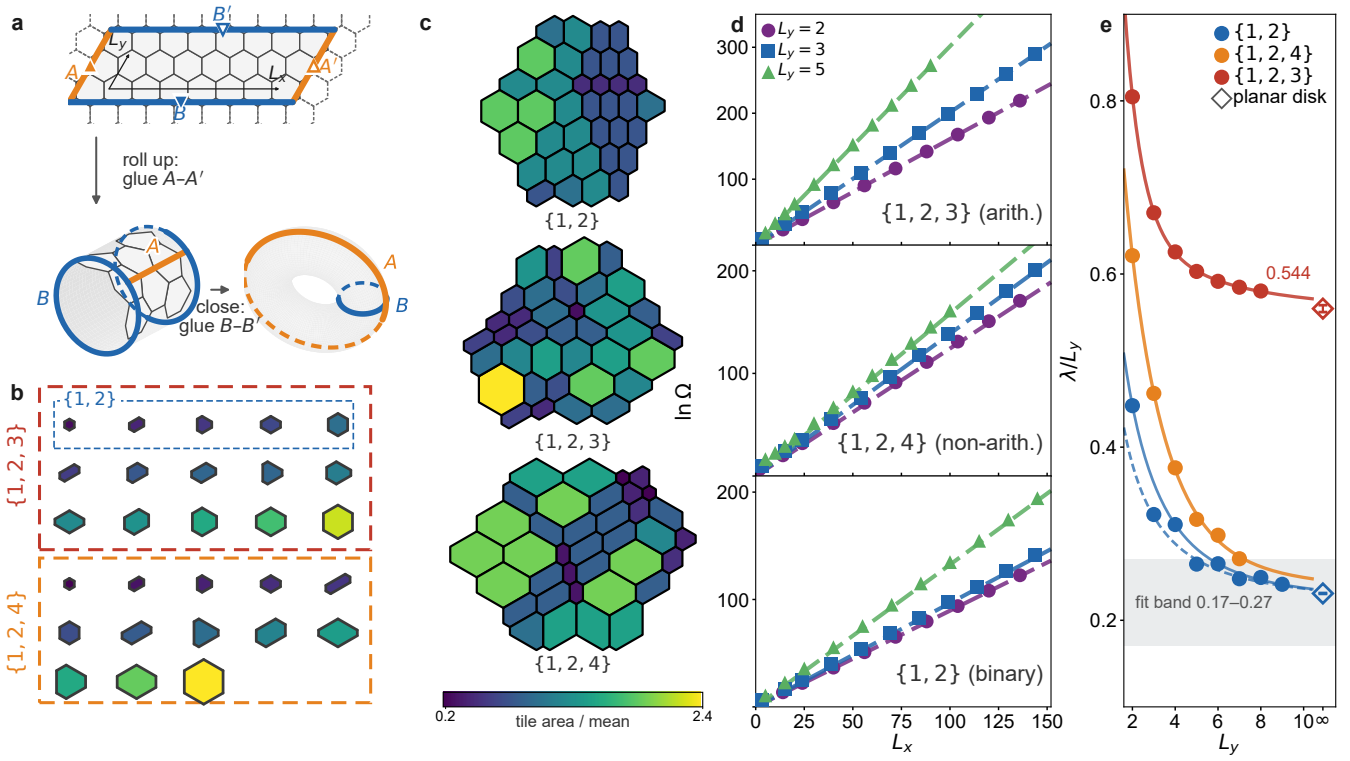


FIG. 3. The honeycomb geometric limit. **a**, The periodic counting domain ($L_x = 8$, $L_y = 3$ drawn): opposite sides identified in pairs (A–A′, B–B′), giving a torus on which every edge is shared by exactly two hexagons; the counting data use the sheared fundamental domain (SI S6.4). **b**, Tile libraries as equiangular hexagons (edge length = value; one representative per dihedral orbit): the 15 classes of {1,2,3}, with the 5 sub-classes of {1,2} boxed, and the 13 classes of {1,2,4}. **c**, Sample patches for the three shape classes, coloured by tile area relative to the mean. **d**, Torus counts, $\ln \Omega$ against L_x, stacked by shape class (top to bottom: {1,2,3}, {1,2,4}, {1,2}) and widths $L_y = 2$, 3, 5: the counts at the resonant sizes (symbols) sit on the transfer-matrix asymptotes (dashed); counting is read on these sizes (SI S6.3). **e**, Width scaling of the per-column entropy current: the points are the exact Perron rates $\lambda(L_y)$ of the strip transfer operator; the curves are the mixed-model fits whose $W \to \infty$ limits are the labelled asymptotes ($s = 0.544$ for the arithmetic class; the binary and non-arithmetic curves fall together at 0.2 inside the shaded fit band 0.17–0.27); open diamonds mark the direct planar-disk count at the ∞ position (binary 0.231, arithmetic 0.560; error bars within the symbols; SI S6.3).

assignment, and each spot carries an anisotropic diffuse halo (Fig. 4a), the metric field's fluctuation signature and the direct analogue of the phason diffuse scattering of fixed-shape random tilings.[11–13] Counting and diffraction are complementary instruments; the subsections below measure their division of labour.

The translation ladder: entropy-blind, halo-visible

Entropy is translation-invariant; diffraction is not. Adding a constant to every edge length maps solutions to solutions bijectively (the closure conditions are homogeneous; Lemma 2, SI S7.3), so $\Omega(\{1,2,3\}) = \Omega(\{4,5,6\})$ pointwise and the entropy density is the same $s_\infty = 0.544$: the affine shape class cannot tell the two domains apart. The diffraction can. Along the ladder $\{1,2,3\} \to \{2,3,4\} \to \{3,4,5\} \to \{4,5,6\} \to \{5,6,7\}$, at fixed network, size and protocol, the coherent fraction of the second-shell intensity (the strong {11}-type $\sqrt{3}$ shell, $|G| = 4\pi\sqrt{3}/3$), $f_{\mathrm{coh}} = |\langle F\rangle|^2/H_{\mathrm{ref}}$, climbs from 9×10^{-4} to 0.58, monotonically as the relative dispersion σ/μ falls from 0.41 to 0.10, while the halo-to-core ratio falls from 17 to 0.6 against its flat-field value of 25 (Fig. 4b; the two smoothest rungs are the ladder extension of SI S9). The counting classification forgets the offset; the halo remembers it. The translation is, for the metric field, an exact amplitude flow: in the mean-bond normalisation used throughout, every displacement field scales by a single factor $\alpha = \bar{\ell}/(\bar{\ell} + c)$ (verified to machine precision; SI S10.4). The ladder therefore changes the *size* of the metric fluctuation, never its structure, which is why the coherent fraction follows the Debye–Waller law of the flow, $f_{\mathrm{coh}} = \exp[-|G|^2(\sigma/\mu)^2]$, with no free parameter: 1% at the two smoothest rungs and 25% or better across the arithmetic ladder (SI S10.4). At the disordered end the readout is patch-limited: the coherent fraction falls with patch size, by a factor of four from

127 to 271 holes, approaching the law from above (SI S10.5). Both base classes sit deep in the disorder regime (coherent fractions 1.8×10^{-4} for {1,2} and 9.0×10^{-4} for {1,2,3} at $N = 169$; halo-to-core ratios 19 and 17); the binary domain's own runs differ by a factor of 56 (SI S9), the thinnest class settling slowly into distinct roughness sectors. The arithmetic ladder has no such ambiguity: its rungs are one sample relabelled by the translation, and its readings reproduce exactly, window by window (SI S9).

Halos and the displacement spectrum: what the diffraction measures

What the halo reads out beyond intensity is the anisotropy of the metric field's fluctuation spectrum. The displacement field of each configuration against the ideal reference decomposes, per wavevector, into longitudinal and transverse parts in the phason convention of Socolar et al.,[33] and the spectrum ratio $S_\perp/S_\parallel$ measures the anisotropy directly. Across the six exact-uniform domains the halo-window ratio, resolved over the ten bins $k = 0.34$–1.06 that carry resolved box modes (SI S10.3), sits at 0.14–0.17 (0.136 ± 0.049 on every arithmetic domain and 0.169 ± 0.056 for the binary), drifting downward across the window by a factor of 2.7 (slope -0.13 ± 0.05). The ratio is exactly invariant along the ladder, because a ratio of two components at one wavevector cannot see the global rescaling that swings the coherence; the binary class, a different constraint set, carries its own ratio. The halo anisotropy r, the second-moment aspect of each spot's diffuse cloud, lies below unity for every domain ($r = 0.29$–0.40; Fig. 4c), the radial elongation of a longitudinal-dominated spectrum. In real space the same ensembles complete the picture: the roughness w_{rms} grows far slower than linearly with patch size, from 0.33 to 0.39 ({1,2}), 0.41 to 0.53 ({1,2,3}) and 0.16 to 0.21 ({4,5,6}) between $R = 3$ and $R = 10$ (Fig. 4d), equally consistent with a logarithmic and a shallow power law (SI S10.1).

Metric tension: the thermodynamics of the counted variable

The metric random tiling has a thermodynamics beyond its uniform ensemble. Weighting every configuration by $\exp(-h\sum_e x_e)$, a tension rewarding compressed ($h > 0$) or stretched ($h < 0$) edge lengths, defines the conjugate family π_h, sampled exactly by the same machinery (SI S9; representative biased configurations in Fig. 4e). The mean length density $m(h) = \langle\sum l\rangle_h/F$ is the metric equation of state: it falls monotonically from 8.91 to 3.10 per face between $h = -0.30$ and $+0.30$ (Fig. 4f), and the entropy cost follows by integration over the measured tensions, $\Delta s(h) = \int_0^h h'm'(h')\,dh'$: -0.490 and $-0.488\,k_B$ per face at the two strongest tensions, symmetric to the ensemble's precision, as the exact mirror symmetry requires (the identity $m(h) + m(-h) = 4$ per edge holds to 0.4σ; SI S9). Where the fixed-shape ensembles tension the *arrangement*,[7] the metric ensemble tensions the *lengths themselves*: the tension couples directly to the counted variable, and the entropy cost is the quantitative price of deforming the disorder.

Replacing every length l by $4 - l$ maps the $+h$ family onto the $-h$ family exactly (SI S10.4), so each tension is read twice: directly, and again through the mirror ensemble, an exact-distribution sample of the same measure. The two readings agree where the response is large: at the stretched corner $f_{\mathrm{coh}} = 0.55$ and 0.54 from the two samplings, and at the compressed corner 0.18 against 0.20, both far above the uniform value 9×10^{-4}, while the halo weight collapses below the flat-field level. The strong tensions therefore drive the ensemble to near-uniform corners ($m = 8.91$ and 3.10 against 6.01 per face) and sharpen the beam. At intermediate tension the coherence dips below its uniform value at $h = +0.15$ ($1.2\times10^{-4}\pm0.7\times10^{-4}$) and rises above it at $h = -0.15$ ($4.3\times10^{-3}\pm2.9\times10^{-3}$), the ragged mixture regime of the value simplex: the response is non-monotone in the tension.

DISCUSSION

Two general principles follow from the classification. The first is that metric entropy belongs to the closure system rather than to the network: whether a tiling can carry a countable metric disorder is decided by how the constraint kernel scales and by the arithmetic of the value domain and of the counting domain, so the verdict for a new network has to be computed rather than inferred. The second is that counting and diffraction are not interchangeable instruments. On one and the same ensemble, the count is exactly blind to an axis of the value domain that the halo resolves, while the displacement anisotropy is invariant along it; the consequence is practical as well as conceptual: it says which measurement sees what.

Both principles open concrete programmes. The classification would become predictive with two theorems: one for when every face remains an independent module, and one for when the integer points of a constraint space are thick rather than thin. The constructive side of the second is now settled for the four carriers themselves (SI S8.3); the criterion that would predict it on a new network is not. The two constructions suggest a dichotomy, that local flips exist precisely when the alphabet contains a three-term arithmetic progression, which would read the closed-form arithmetic of Sec. from the alphabet side. The counting domain poses its own questions, from the asymptote of the 4.6.12 rates to a general

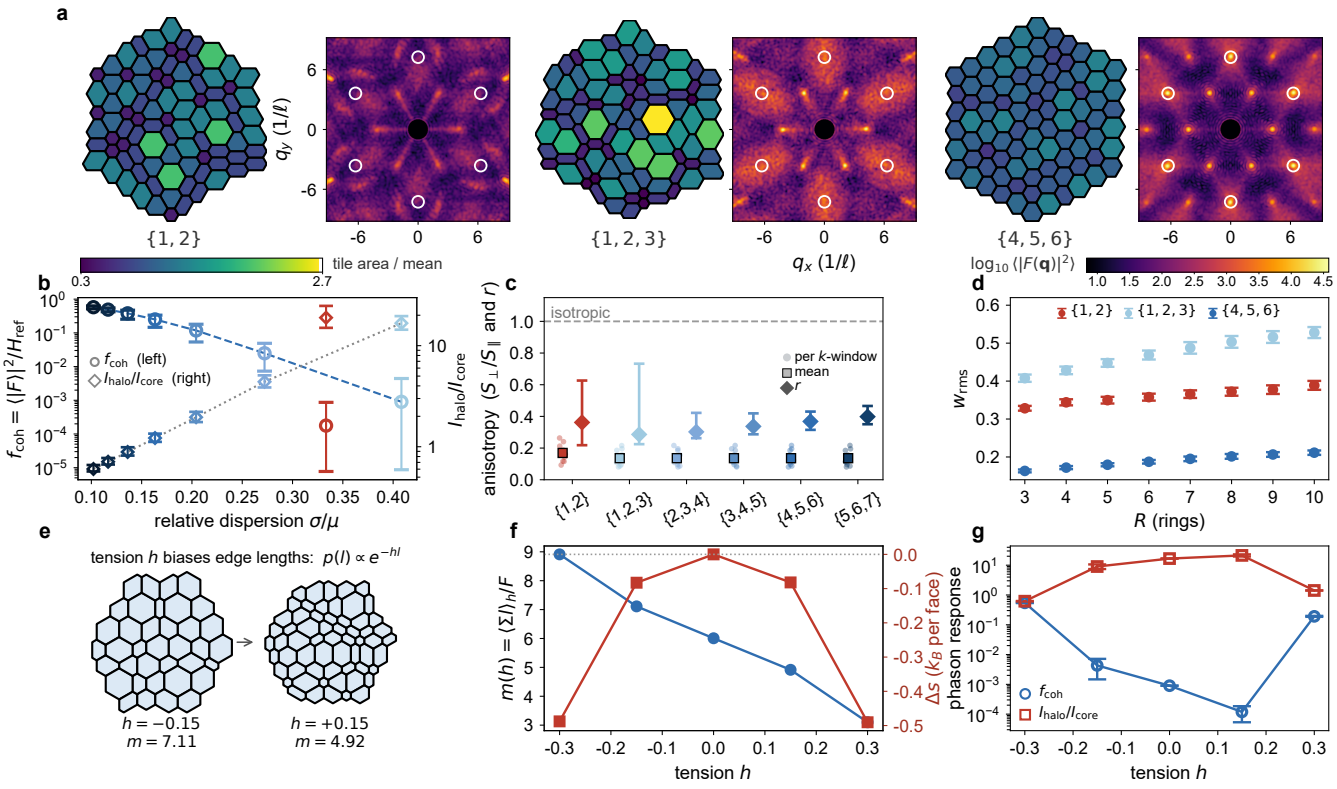


FIG. 4. The value domain in diffraction: halos, anisotropy, and metric tension. **a**, Real-space patches and ensemble-averaged $|F(q)|^2$ for three value domains (20 configurations each; log colour scale; white circles, second-shell positions of the ideal-patch reference, the strong {11}-type $\sqrt{3}$ shell at $|G| = 4\pi\sqrt{3}/3$; the weak {10}-type first shell at $|G| = 4\pi/3$ appears as the inner hexagon of features rotated by 30°, degrading into radial streaks at the disordered end). **b**, The translation ladder: coherent fraction f_{coh} (left axis, open circles; blue, the arithmetic domains $\{n, n{+}1, n{+}2\}$, n = 1–7; red, the binary {1,2} domain off the guide) and halo-to-core ratio (right axis, open diamonds; flat-field value 25) against the relative dispersion σ/μ; bars, configuration-bootstrap intervals (the intervals of the smoothest rungs are smaller than their markers). **c**, Anisotropy per domain: the spectrum ratio $S_\perp/S_\parallel$ per k-window (dots) with window means (squares; 0.136 ± 0.049 arithmetic, 0.169 ± 0.056 binary) and the halo aspect r (diamonds); dashed line, isotropic. **d**, Real-space displacement roughness $w_{rms}(R)$ against patch radius R. **e**–**g**, Metric tension on {1,2,3}: **e**, biased configurations at $h = \mp 0.15$; **f**, equation of state $m(h)$ (left axis) and entropy cost $\Delta s(h)$ (right axis); **g**, diffraction response $f_{coh}(h)$ and $I_{halo}/I_{core}(h)$, non-monotone in h (markers, mean of the two mirrored samplings of each tension; bars, their difference).

account of when a sub-law is transient and when it is the physics, for which the planar counts of Fig. 3 provide a first calibration. The elasticity side is the next target: a strain theory of discrete-metric tilings that would predict the measured displacement anisotropy; whether the stiffness defined by such a theory follows from the count, as it does in fixed-shape ensembles, or stands beside it, is open.

The neighbours of this row are being built from the other side. Correlated-disorder programmes have produced crystals with occupational, positional and orientational disorder, and frameworks whose connectivity is rewired while the lengths stay fixed;[34–37] Truchet architectures realised in molecular networks have become quantitative platforms for disorder physics,[38, 39] simulation itself has begun to expand the vocabulary of order beyond the crystal paradigm,[40] and the metric row is the one in which the fluctuating variable is the edge length itself. The entropy counted here is the fixed-composition one: fixing the counts of the length values removes alloy rearrangements, and the remainder is an entropy channel of its own, blind to the mixture and carrying the geometry (SI S11, Fig. S11). The diffraction fingerprints of Sec. , Bragg spots whose coherent fraction and halo weight track the value domain, are the observables by which such a disorder, countable and crystal-like at once,[41] would be recognised, and they carry two discriminants against its neighbours: the bond-angle distribution stays exactly at the lattice angles for any closure-respecting assignment ($\psi_6 \equiv 1$), unlike displacive disorder, and a metric mixture on one vertex type gives one set of Bragg spots wrapped in metric halos, not the two spot sets of a phase-separated mixture.

In 1619, Kepler drew the eleven uniform tilings in the

Harmonies Mundi: figures of harmonic proportion, in which disorder appears nowhere, and could not yet be asked about. The answer, four centuries on, has the form Kepler would have asked for: not a dimension but a count, seven zeros and four positive carriers, carried by a disorder the drawings never showed, living in winding sectors, and readable in the halos of a diffraction pattern.

DATA AVAILABILITY

All counting CSVs and structure-factor outputs are openly available in the companion repository https://github.com/shtusynelab/metric-random-tiling (complete file map in the Reproducibility Supplement, Table R2), archived on Zenodo (DOI: 10.5281/zenodo.24156134, concept DOI).

CODE AVAILABILITY

All counting, figure-generation, and validation scripts cited in the Reproducibility Supplement (Table R2), with per-script manifests, run logs, and per-configuration seeds, are openly available in the companion repository https://github.com/shtusynelab/metric-random-tiling, archived on Zenodo (DOI: 10.5281/zenodo.24156134, concept DOI).

AUTHOR CONTRIBUTIONS

X.W. performed the research, developed the counting and analysis pipeline, and wrote the manuscript; Q.J. conceived and supervised the research and edited the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

Methods

Model and exact arithmetic. Networks are the edge-to-edge planar uniform tilings (vertex configuration notation as in Table S3/S2) plus the Penrose rhombi; honeycomb patches and tori are built with exact $\mathbb{Q}(\sqrt{d})$ arithmetic ($d = 2$, 3) so that closure is decided exactly; three independent rank verifications (Fraction, two modular primes, float SVD) guard every matrix reduction. The equiangular hexagon closure conditions reduce to two linear equations per face.[30]

Counting. All entropy densities are computed on tori, the parallelogram counting domain with both pairs of opposite sides identified (Fig. 3a): every edge is then shared by exactly two faces, and no boundary degrees of freedom remain. The convergence of $\ln \Omega/F$ to the bulk value is governed by the arithmetic of the counting domain: finite sizes are either coprime, where the exact sub-law of Sec. can mask the entropy, or non-coprime, where the winding sectors carry it, so the entropy density is read along fixed-width families, $s_\infty = \lim_{L_y\to\infty} \lim_{L_x\to\infty} \ln \Omega/F$, with the crossover scale $L^*(L_y)$ growing rapidly with width (SI S6.3); the construction is specified in full, with worked smallest-torus verifications, in SI S1.2. The open-boundary comparisons use a simply connected patch count: a hexagonal-envelope disk of R rings ($N = 3R(R + 1) + 1$ faces) is counted exactly by a column-recursion frontier DP with dense 128-bit accumulators ($R = 3$–7 for the binary and arithmetic classes), anchored to full and blocked enumeration at small sizes (SI S6.3).

Conventions. Every fundamental-domain cell of the counting torus contains exactly one hexagonal face, so $F = L_x \cdot L_y$ and the per-tile entropy at fixed width is $s = \lambda/L_y$ (λ the per-column winding flux); "tile" = face = hole throughout; edge-length alphabets are labelled from 1 throughout ({1,2}, {1,2,3}, {1,2,4}); all counts are invariant under integer translation of the alphabet (an affine shape-class property), so the counting scripts may realise the same shape classes with equivalent integer labels.

Counting runs on a three-tier ladder, each tier anchored in the one below (engines, reach, and verification: SI S11). (i) *Exact enumeration*: DFS over partial assignments with integer kernel parametrisation, fail-closed allocation budgets, and brute-force cross-checks on the smallest systems. (ii) *Transfer matrices*[11]: a keyed dynamic program over partial assignments (states merged by key before vectorisation, since path counts are not state counts) builds a single-column strip operator T; strips are counted by iterating T and tori by the exact trace $\Omega(L_x \times L_y) = \mathrm{Tr}(T^{L_x})$; at scale the sparse trace runs in checkpointed chunks, systems with face-based constraints (4.6.12) are counted by constraint-space elimination and keyed frontier dynamic programs over face-closure orderings, and closed forms replace counting where theorems provide them (verified bitwise at every measured size). (iii) *Calibrated stochastic estimators*, deployed only where exact counts cap out: Knuth-type unbiased tree sampling[42] (entropy-density bias ≈ 0.01–0.02 after calibration against exact anchors) and Hutchinson stochastic traces[43] (exactly unbiased by the trace identity; validated at known-answer sizes before deployment). Every stochastic value sits against exact values at overlapping sizes, and expensive extrapolations are guarded by registered out-of-sample predictions.

The split-row carriers' entropy signs are decided by cylinder transfer-matrix spectral radii at the matched

two-value alphabet: the asymptotic per-column growth rate, immune to the finite-length transients that mimic perimeter laws in short-window counts. The sign is carried to richer alphabets by an alphabet-monotonicity lemma, and every stochastic link is calibrated against exact anchors (SI S8.2).

The honeycomb entropy densities of Fig. 3d,e are width-asymptotes of transfer-matrix spectral radii $\lambda(L_y)$ (binary L_y = 2–9, non-arithmetic 2–7, arithmetic 2–8), extracted under a four-gate discipline (machine-bitwise anchors, head-side purity, tail connectivity, known-answer calibration; SI S11) and fitted with the mixed global form $\lambda(L_y)/L_y = s + c/L_y + A'q^{L_y}$; the fit pins the arithmetic class at $s = 0.544$ and places the binary and non-arithmetic classes in a common band 0.17–0.27 (SI S6.2). Long-chain exact counts (L_x up to 150, 1 100+ points) anchor the counting side; the non-resonant sizes (all coprime at these widths) of the binary and non-arithmetic classes follow the exact perimeter law bitwise (SI S6.2–S6.3).

Constructive lower bounds. The positivity of the four carriers is anchored by two explicit mechanisms: completion-product certificates (a finite patch carrying two admissible completions at equal boundary values multiplies the count by two per disjoint copy; ternary alphabet, all four carriers), and, on the binary honeycomb, a stripe-compatible site construction whose every subset admits a completion, $\Omega \geq 2^{(2/25)LW-O(W)}$ on every strip of width $W \geq 3$. Certificates are finite objects verified by exact machine checks (dual-algorithm counting, per-width construction checks, injectivity and end-to-end count tests, and negative controls on zero-entropy systems); statements and proofs are collected in SI S8.3, with scripts and verification records in the Reproducibility Supplement.

Diffraction. Ensembles: twenty independent closure-respecting configurations per value domain at $N = 169$ holes (six domains: the two base classes and the four ladder offsets; SI S9, Table S7), generated by the two-stage exact-uniform sampler (difference-coordinate structure theorem with parallel tempering; SI S9) with per-configuration closure verification; the four ladder offsets are the same sampled configurations under the exact translation of the alphabet (Sec. , SI S9). Structure factors are exact sums over the unit-cell contents of each configuration with a convex-hull mask; the coherent fraction f_{coh} uses the ensemble-averaged complex amplitude at the six second-shell spot positions of the ideal-patch reference on the same grid (the strong {11}-type $\sqrt{3}$ shell, $|G| = 4\pi\sqrt{3}/3$ in bond-length units; positions derived from the ideal patch's structure factor, SI S9), measured against the ideal-patch reference H_{ref} in the same estimator; halo metrics integrate the annulus 0.22–1.10 around each spot against the core (≤ 0.22); the displacement field is matched to the ideal reference site-by-site through the graph topology and decomposed into longitudinal and transverse spectra (SI S10.3); the ladder extension to {7,8,9} and the patch-size series (hexagonal discs of 127–331 faces on the 21×21 ensembles) are specified in SI S9 and S10.5. Configuration-bootstrap intervals ($B = 200$); all seeds and per-run parameters are archived with the raw ensembles listed in the Reproducibility Supplement (Table R2).

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